

Lecture 1 — Obesity: Companion Figures

Prof. Kevin Murphy, 22 Oct 2025

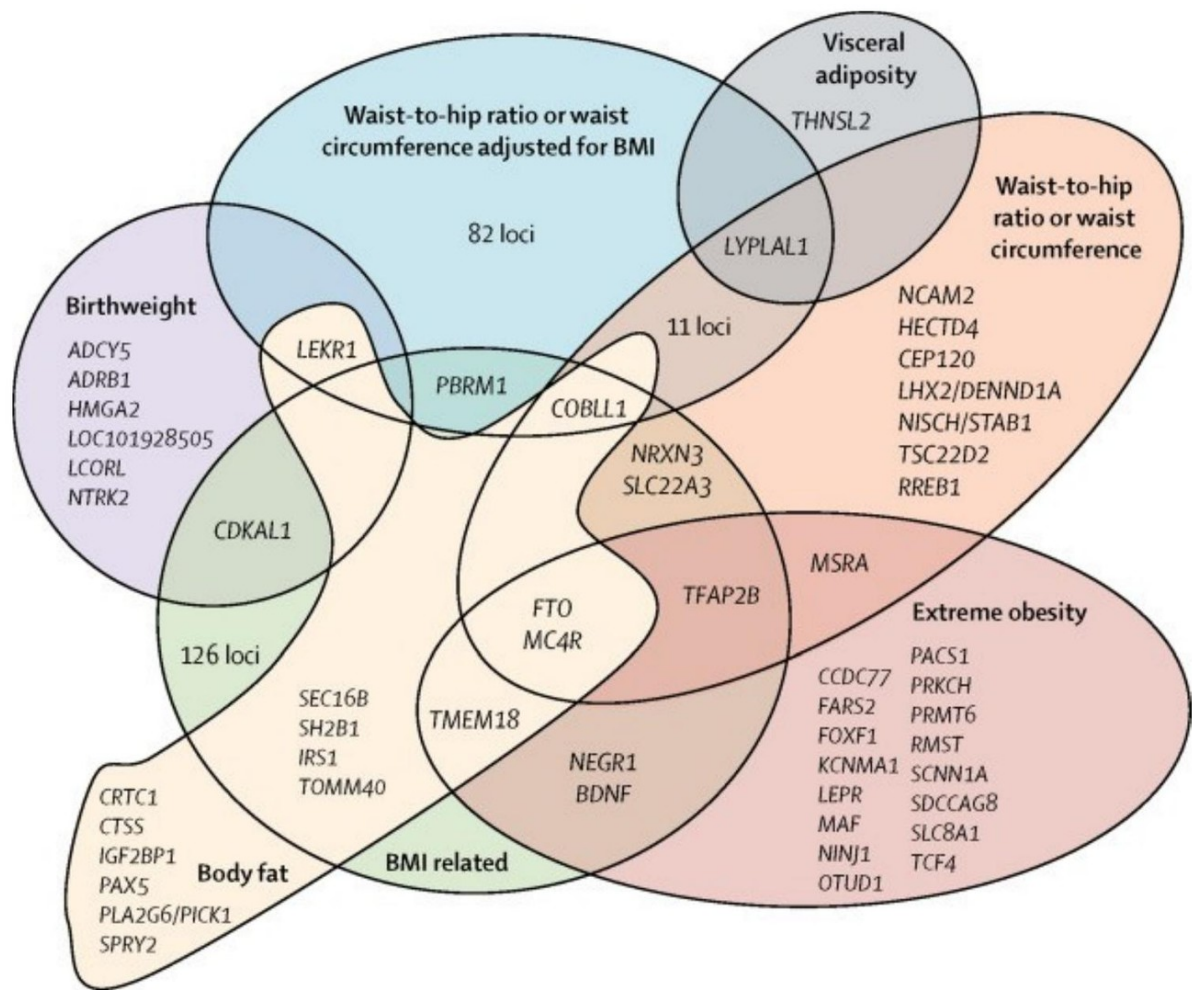
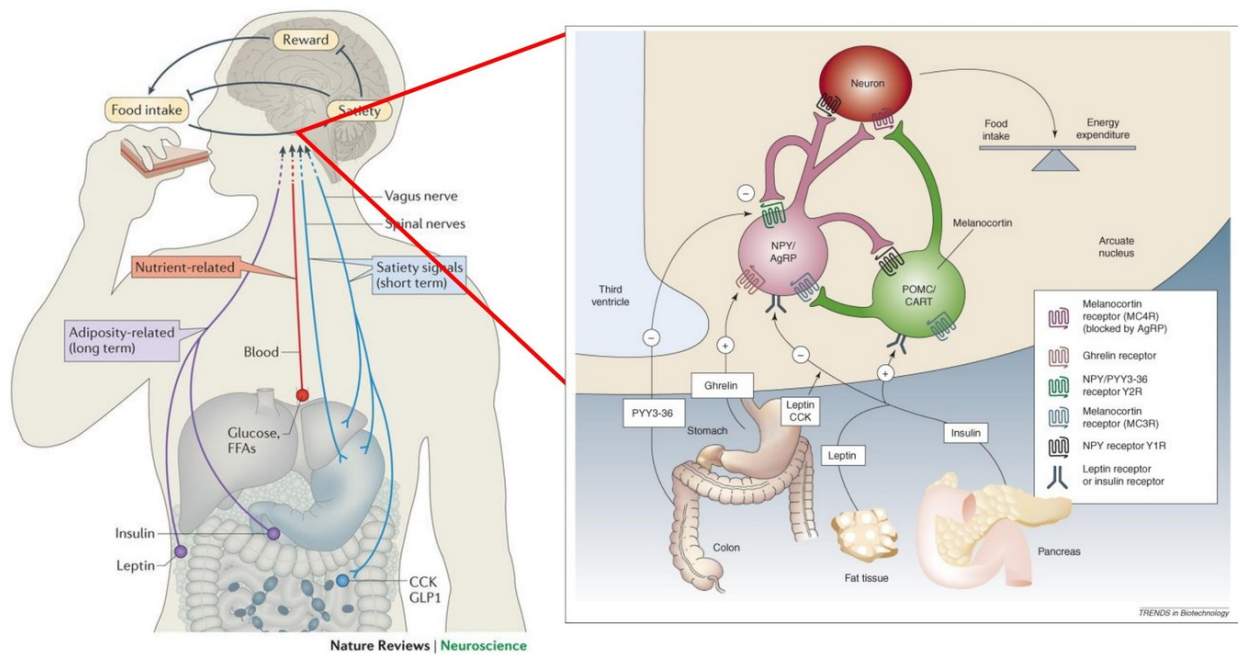


Figure L1.1 — Genome-wide association loci grouped by obesity-related trait (BMI, waist-to-hip ratio, birthweight, extreme obesity). Most loci are trait-specific; few overlap. (Goodarzi, Lancet Diabetes Endocrinol 2018)



Morton GJ, Meek TH, Schwartz MW. Neurobiology of food intake in health and disease. *Nat Rev Neurosci.* 2014 Jun;15(6):367-78.

Stocker CJ, Cawthorne MA. The influence of leptin on early life programming of obesity. *Trends Biotechnol.* 2008 Oct;26(10):545-51.

Figure L1.2 — Hypothalamic regulation of appetite. Left: peripheral signals (insulin, leptin, gut hormones, vagal afferents) reaching the brain. Right: arcuate nucleus circuits — NPY/AgRP (orexigenic) inhibit POMC/CART (anorexigenic) via MC4R signalling.

All Causes of Mortality

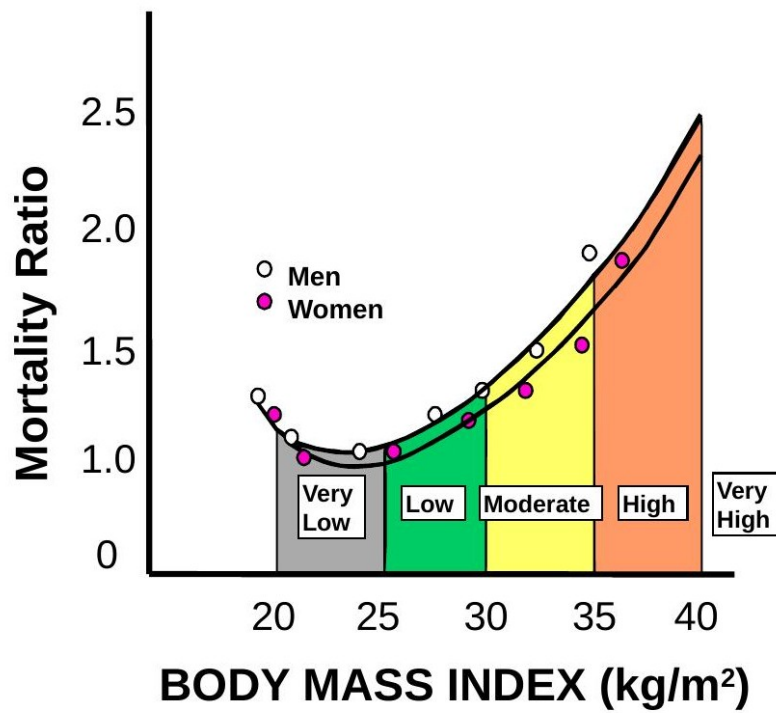
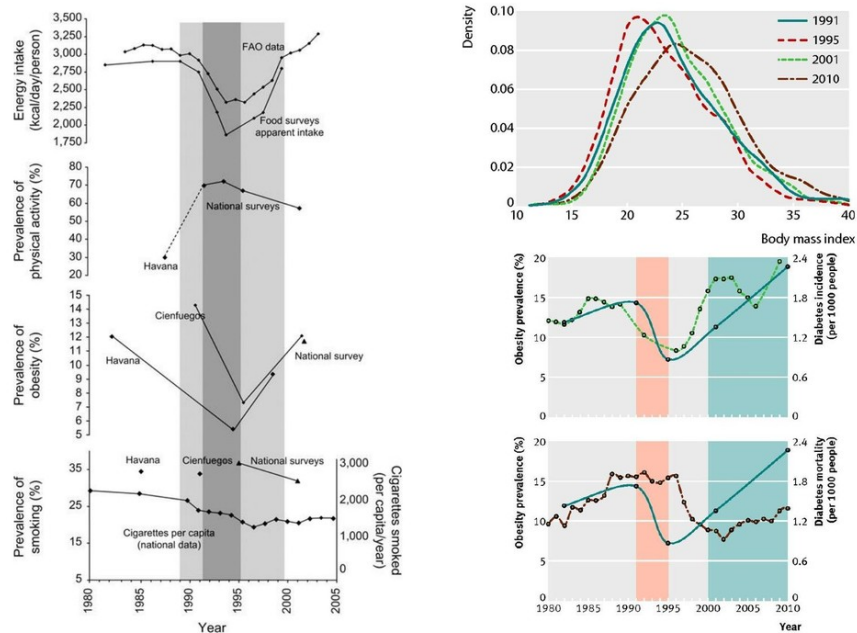


Figure L1.3 — U-shaped relationship between BMI and all-cause mortality ratio. Nadir at BMI 20–25; rises at very low BMI and steeply at BMI ≥ 30 . Mortality ratio approx 2× at the morbidly obese end.

What happened in Cuba after obesity levels

weight loss and
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J, Orduñez P,
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Population-wide weight loss and
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and cardiovascular mortality in C
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secular trends. Franco M, Bilal U
Orduñez P, Benet M, Morejón A,
Caballero B, Kennelly JF, Coope
BMJ. 2013 Apr 9;346:f1515. doi:
10.1136/bmj.f1515.

Figure L1.4 — Cuba 1980–2010: post-Soviet economic crisis caused population-wide weight loss; obesity prevalence and diabetes incidence/mortality fell concordantly, then rose again as economy recovered. (Franco et al, BMJ 2013)

Leptin necessary for normal physiological function

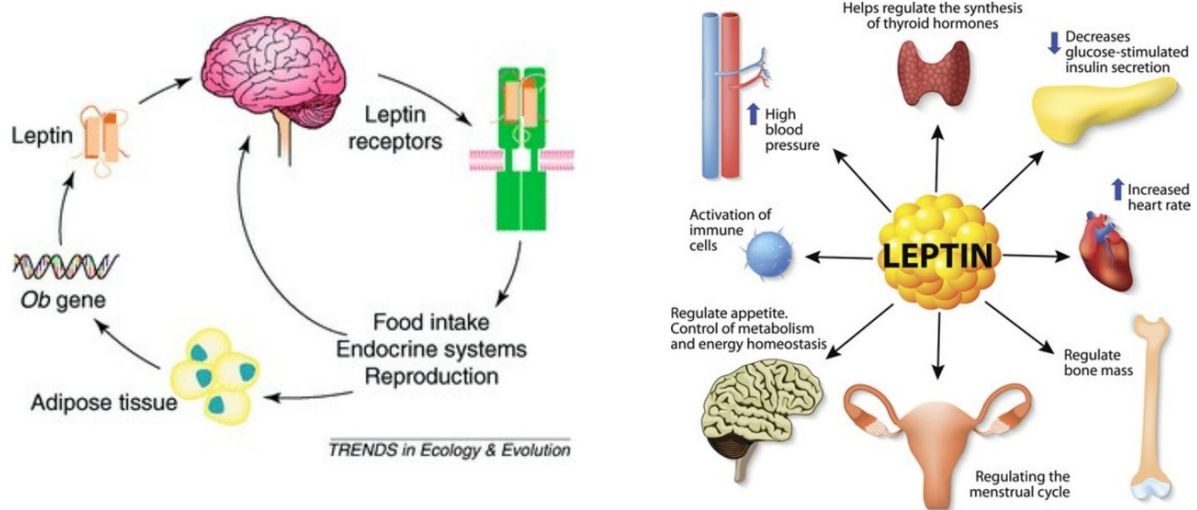
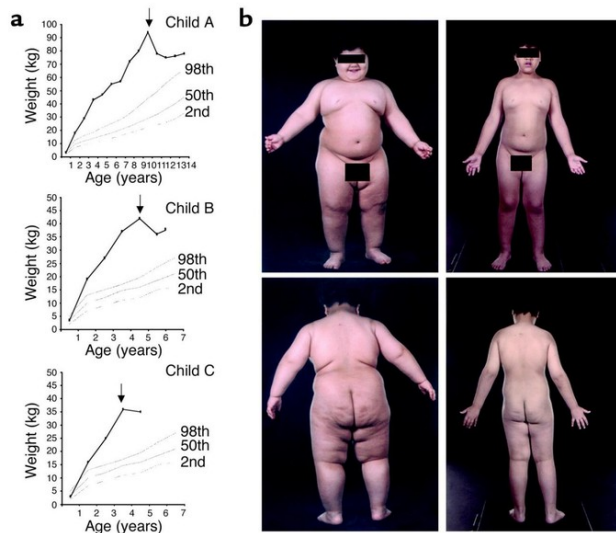


Figure L1.5 — Leptin: secreted from adipose tissue, signals to hypothalamic leptin receptors, with multi-organ actions on appetite, metabolism, reproduction, thyroid synthesis, immunity, BP, heart rate and bone mass.

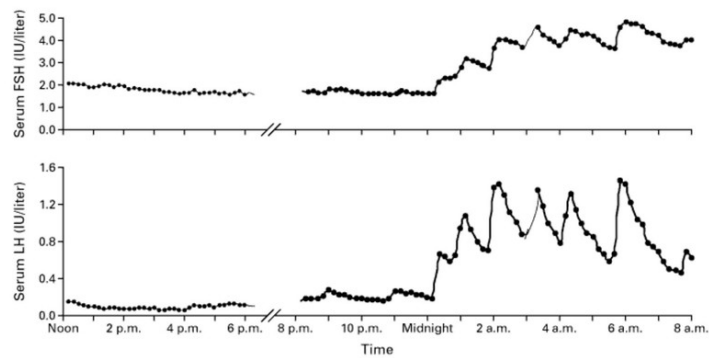


- Body weight in leptin deficient children receiving recombinant leptin therapy.

Farooqi et al J Clin Invest. 2002 October 15;110(8):1093-1103.

Endocrine disorders: Describe the clinical features and treatment options of endocrine disorders.

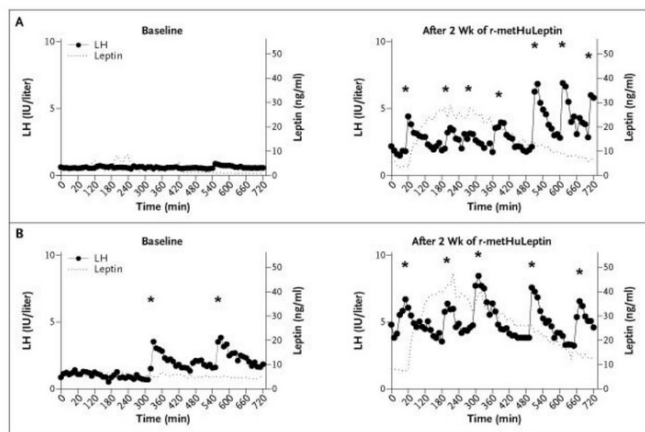
Figure L1.6 — Recombinant leptin therapy in three congenitally leptin-deficient children. Weight-for-age trajectories drop sharply after starting leptin (arrows). Photographs: before vs during treatment. (Farooqi et al, J Clin Invest 2002)



- Serum Follicle-Stimulating Hormone (FSH) and Luteinizing Hormone (LH) Concentrations in a Patient with Congenital Leptin Deficiency after 12 Months of Leptin Treatment.

Effects of recombinant leptin therapy in a child with congenital leptin deficiency.
Farooqi IS, Jebb SA, Langmack G, Lawrence E, Cheetham CH, Prentice AM, Hughes IA,
McCamish MA, O'Rahilly S. N Engl J Med. 1999 Sep 16;341(12):913-5.

Figure L1.7 — In a leptin-deficient patient, baseline FSH and LH appear non-pulsatile. After 12 months of recombinant leptin therapy, normal nocturnal pulsatility returns — restoring HPG-axis function. (Farooqi et al, NEJM 1999)



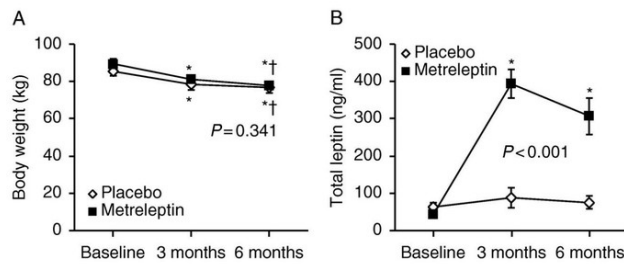
- Representative Patterns of LH Pulsatile Characteristics at Baseline and after Two Weeks of Leptin Therapy.

Recombinant human leptin in women with hypothalamic amenorrhea. Welt CK, Chan JL, Bullen J, Murphy R, Smith P, DePaoli AM, Karalis A, Mantzoros CS. N Engl J Med. 2004 Sep 2;351(10):987-97.

Figure L1.8 — Women with hypothalamic amenorrhoea: baseline LH pulses absent or markedly blunted; after 2 weeks of r-metHuLeptin, LH pulsatility is restored. (Welt et al, NEJM 2004)



Leptin resistance



Leptin administration to overweight and obese subjects for 6 months increases free leptin concentrations but does not alter circulating hormones of the thyroid and IGF axes during weight loss induced by a mild hypocaloric diet.

Shetty GK, Matarese G, Magkos F, Moon HS, Liu X, Brennan AM, Mylvaganam G, Sykourti D, Depaoli AM, Mantzoros CS. Eur J Endocrinol. 2011 Aug;165(2):249-54.

- Absence of leptin has profound effects, including hyperphagia, lowered energy expenditure, sterility.
- However, leptin is an anti-starvation hormone rather than anti-obesity hormone.
- Presence of leptin is a signal to the CNS that system has sufficient fat reserves for normal functioning- but high leptin has little effect.

Endocrine Disorders: Description, Clinical Features, and Treatment Options of Endocrine Disorders

Figure L1.9 — Metreleptin in overweight/obese subjects raises serum leptin markedly (right) but does NOT produce weight loss vs placebo (left). Demonstrates leptin resistance in common obesity. (Shetty et al, Eur J Endocrinol 2011)

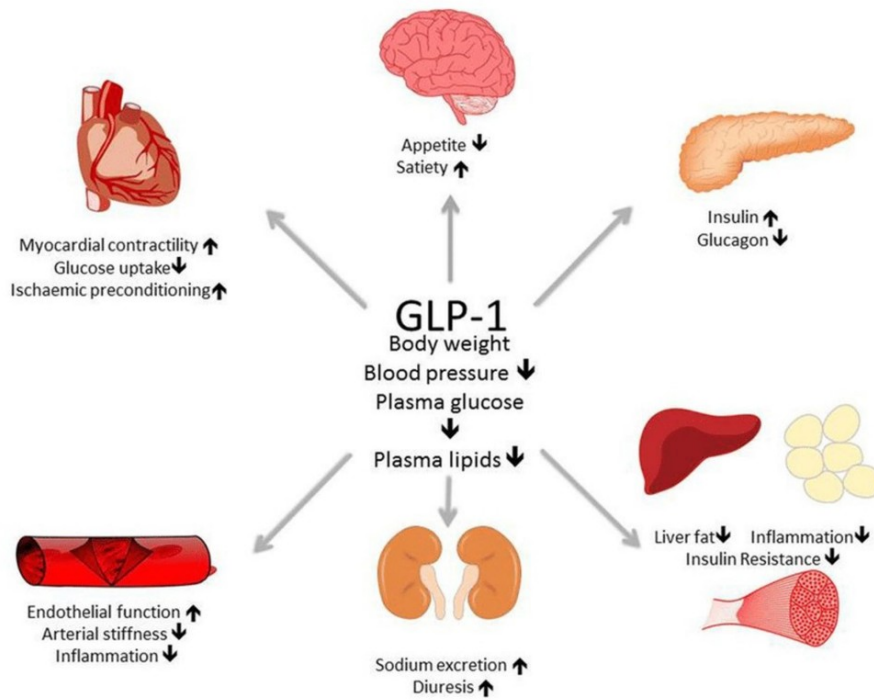


Figure L1.10 — GLP-1 actions across organs: brain (↓appetite, ↑satiety), pancreas (↑insulin, ↓glucagon), heart (↑contractility), kidney (↑natriuresis), vasculature (↑endothelial function), and liver/muscle/fat (↓steatosis, ↓insulin resistance, ↓inflammation).

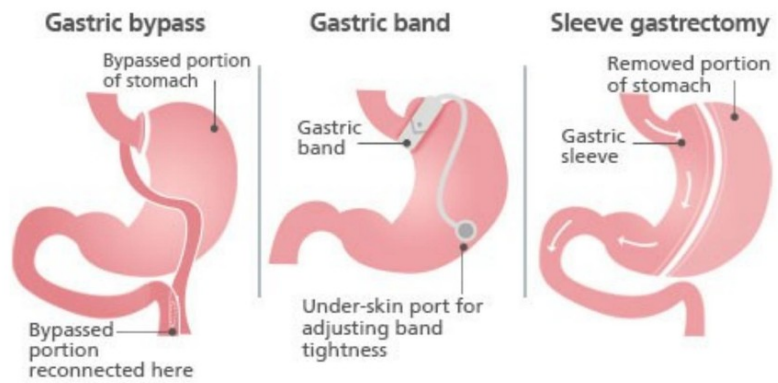


Common types of bariatric surgery

Gastric bypass – the top part of your stomach is joined to the small intestine, so you feel fuller sooner and do not absorb as many calories from food

Gastric band – a band is placed around your stomach, so you do not need to eat as much to feel full

Sleeve gastrectomy – some of your stomach is removed, so you cannot eat as much as you could before and you'll feel full sooner.



Endocrine disorders: Describe the clinical features and treatment options of endocrine disorders.

Figure L1.11 — Three common bariatric procedures. Gastric bypass: small upper-stomach pouch joined directly to small intestine (restriction + malabsorption). Gastric band: adjustable band creates restriction only. Sleeve gastrectomy: stomach reduced to a tubular sleeve (restriction + reduced ghrelin).

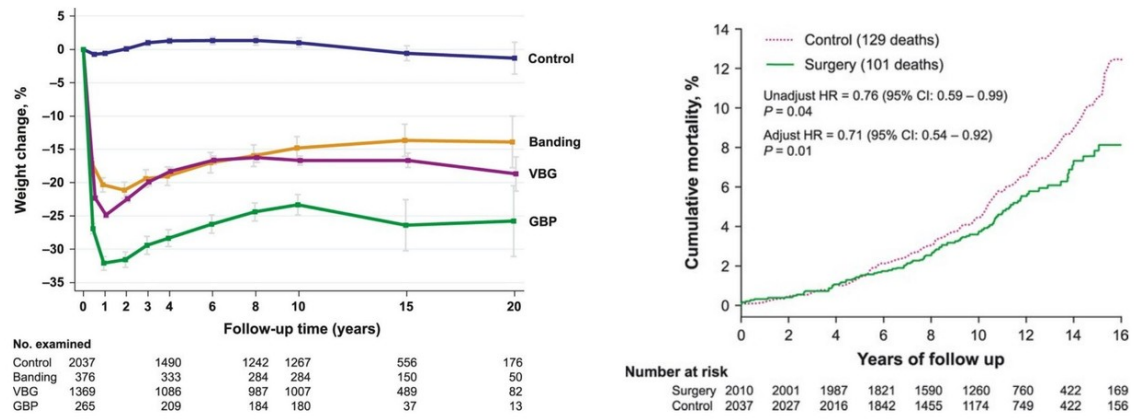


Figure L1.12 — Swedish Obese Subjects study. Left: 20-year weight change by procedure (GBP ~–25%, VBG ~–18%, banding ~–14%, control ~0%). Right: cumulative all-cause mortality lower in surgery group (adjusted HR 0.71, 95% CI 0.54–0.92).